PKM lead 1, $m = 3$, $N = 21$: run report

Runs `M3-N21-t0`, `M3-N21-t1`, 2026-10-18 06:11–10:29 UTC — observations only, no verdict stated here

Sources. Card `backlog/XP-20261015-pkm-m3-n21-m4-n20`, cell `m3-n21`. Protocol `research/pkm_lead1_m3n21_20261016/PROTOCOL.md` (frozen before any $N = 21$ run). Engine `f4_fp_tower` through `examples/pkm_tower_pilot.rs` (`--engine tower`), release build at commit `b7cafe044` (main), binary sha256 `d25eadb1…c265d`, rustc 1.99.1. Commands `run.sh host`, `identity`, `main 0`, `main 1`, `verify`. Records: `runs/host-ip-172-31-35-131….txt`, `runs/identity-N15-t0-ip-….{jsonl,log}`, `runs/M3-N21-t{0,1}.{jsonl,log}`, `progress.txt`, `verify.txt`. Every number below is **recorded** unless labelled **derived**.

1. Host and caps

host	EC2 spot instance `ip-172-31-35-131.us-east-2.compute.internal`, 32 vCPU, MemTotal 129,800,716 KiB (124 GB), no swap; all 32 threads used (`RAYON_NUM_THREADS` unset)
caps (PROTOCOL.md §2, SHARE = 1)	A = 119,416,658 KiB (`ulimit -v`), C = 20,300,000,000 entries (`--max-dense`), budget 345,600 s per target
deviation, disclosed before stage 2	§2 requires $\geq$ 192 GB; this host has 124 GB. The rules for A and C were applied unchanged. Recorded in the host file.
isolation	none claimed: a shape-search sizing probe (one thread, 64 GB cap) ran on the same host throughout

2. Stages

stage	start	wall	recorded outcome
identity ($N = 15$, target 0)	05:54:06	52 s	exit 0. `solving_degree_max` 7, `max_cols` 31,397, `steps` 24, `inconsistent` true, `muladds` 3,185,681,037,100, `max_rows` 52,129, `max_residual_rows` 30,474: every compared field equals round 6′s committed row. `pkm_tower_check verify`: 1 row, 1 agrees.
main, target 0	06:11:43	6,445 s	exit 0. `last_productive_degree` 7, `degree_reached` 8, `inconsistent` false, `oversize` true, `timed_out` true, 27 steps, `max_cols` 667,444, `max_rows` 891,980, `max_dense_entries` 79,330,205,871, `max_nnz` 34,785,087,985, `muladds` 1.560×10^{14}, `x_r` 221,026,520.
main, target 1	08:42:41	6,416 s	exit 0. Identical to target 0 on every field above except `max_nnz` 34,785,087,996, `muladds` 1.560×10^{14} (differs in the 8th digit), `x_r` 704,078,485.
confirmation (§3.3)	—	—	not applicable: no target finished with a D.
verify (§3.4)	10:31	—	1 tower row (the identity row) checked by exhaustive search, 1 agrees; the two $N = 21$ rows carry no verdict to check.

3. The stop

Both targets reduced every degree-7 pair (steps 21–26, `fresh` rows at every step, last finished step 26: 271,278 columns, 430,250 pairs left, all at degree 8) and were then stopped in step 27, the first degree-8 step, before any reduction (`muladds` 0, `residual_rows` 0): the step's reducer block B' would hold 79,330,205,871 entries against C = 20,300,000,000 (`f4_fp_tower` stops as oversize when `dense_entries > max_dense`; the trace prints the generic `oversize: --max-nnz` wording). **Derived:** at 4 bytes per entry that B' is $\approx$ 317 GB, and $\approx$ 460 GB at round 7′s peak-to-B' ratio of 1.45, so the same cap would have stopped a 192 GB host. Peak resident memory before the stop: 67.1 GB (target 0), 67.1 GB (target 1); address-space peak 68.7 GB. The protocol's A and the budget were not reached.

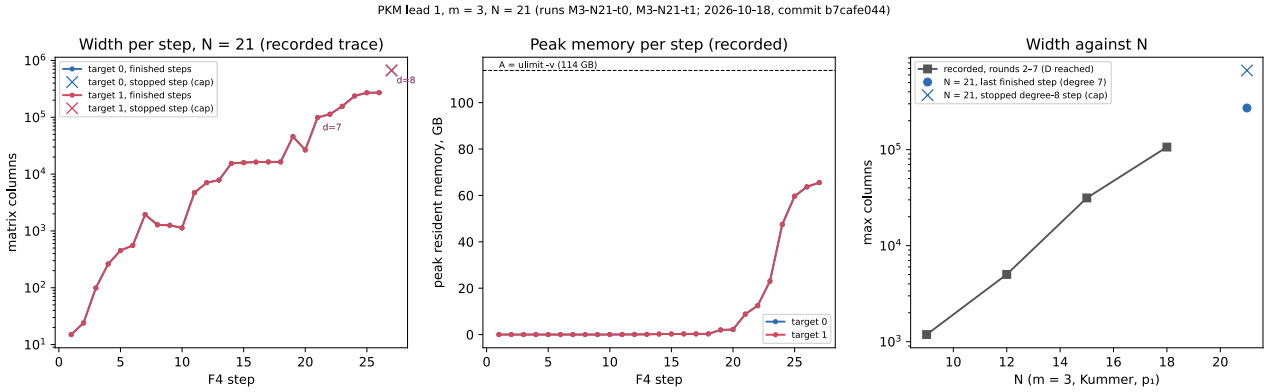


Figure 1: Recorded per-step width and peak memory of both $N = 21$ targets (the two traces coincide), and the recorded width of the $m = 3$ Kummer line at p_1 for $N = 9$–18 (rounds 2–7) beside $N = 21$'s last finished and stopped steps. Runs `M3-N21-t0`, `M3-N21-t1`; source `plot.py`.

4. Against the predictions (PROTOCOL.md §4)

prediction	**recorded**
1. identity holds	holds
2. $D = 7$ on both targets, both refute (low confidence)	not observed: neither target refuted at degree 7; both entered degree 8 and were stopped there by C
3. both targets give the same D, width and step count	holds: identical width, steps and stop
4. width 360,000–475,000, 1.3–1.5 $\times 10^{11}$ B peak, 2.5–7.4 $\times 10^{15}$ muladds (too low if D rises)	last finished width 271,278; stopped step 667,444; peak 67 GB before the stop; 1.56 $\times 10^{14}$ muladds before the stop. The degree-8 step's B' alone exceeds the projected peak by $\approx 2\times$ (derived)
5. 14–60 h on 32 threads	1 h 47 min per target to the stop, which is not a completed system

The reading under §5 is not made here. The rows that may apply are "stopped with $D \geq 7$ only" and "stopped after a productive step at degree ≥ 8"; the recorded trace has every degree-7 pair reduced without refuting and no productive degree-8 reduction. That reading belongs to the result PR.

5. Files

`runs/M3-N21-t0.jsonl`, `runs/M3-N21-t1.jsonl` (one row each plus a summary line), their `.log` step traces, `verify.txt`, `progress.txt`, and `research/large_memory_runs_20261018/EXPERIMENT_LOG.md` for every run on this host.